Components selection reasoning:

1) The 5 V → 3.3 V drop is relatively small and the expected current was unspecified, so a simple LDO was selected for low component count and straightforward implementation.

2) All the discrete components chosen are 0805 package, which I can solder comfortably and even hand etch a single layer pcb. For a mass produced device typically the discrete components sizes can be lowered.

3) Usb type c with 6 pin because the sole purpose was to power the device.

4) For the button and leds the sizes were chosen as for an handheld device.

Design notes:

1) Instead of typical voltage divider network, OP-AMP can buffer the divided sensor voltage, providing a low-impedance source for the ADC

2) With more time I could add a type based debug using usb-uart drivers such as cp2102 however it would add to bom but that would make the product a deliverable one.

3) Low ESR decoupling caps

Optional Stretch

1) Decoupling capacitor is requried to avoid voltage ripples. It aslo absorbs high-frequency noise, and to supply instant current during rapid switching. In pcb design the decoupling capacitor would be placed as close possible to the MCU power pins.

2) Considering human body model I would confirm whether there’s a need for ESD doides with senior engineer. Other suggestions depends on the requirements however I would also confirm with the senior engineer whether a conformal coating is required which provides protection to the pcb from dust, water (sweat), contact.